True or false? (unless otherwise specified.) Questions 1–15: 1 point for a correct answer, 0 otherwise.

Answer grid

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	B
True	☐	☐	☒	☒	☒	☒	☐	☒	☒	☐	☒	☐	☐	☒	☐	☐
False	☒	☒	☐	☐	☐	☐	☒	☐	☐	☒	☐	☒	☒	☐	☒	☒

1. **Degree of precision.** The degree of precision of the following quadrature rule is 3:

$$\int_{-1}^{1} f(x)\,\mathrm{d}x \approx f\left(-\frac{1}{3}\right) + f\left(\frac{1}{3}\right). \tag{1}$$

False. The rule is exact for 1, x and x^3 by symmetry, but for x^2 it gives $\frac{2}{9} \neq \frac{2}{3}$. Its degree of precision is 1.

2. **Degree of precision.** The degree of precision of the following quadrature rule is 1:

$$\int_{-1}^{1} f(x)\,\mathrm{d}x \approx 2f(0) - f(-1) + f(1).$$

False. For $f(x) = x$ the rule gives $2 \cdot 0 + 1 + 1 = 2$, whereas the integral is 0. The degree of precision is 0.

3. **Degree of precision.** The degree of precision of the following rule is 5:

$$\int_{-1}^{1} f(x)\,\mathrm{d}x \approx 2f(0) + \frac{1}{3}f''(0) + \frac{1}{60}f^{(4)}(0).$$

True. By symmetry the rule is exact for odd monomials. It is exact for 1 (2), for x^2 ($\frac{1}{3} \cdot 2 = \frac{2}{3}$) and for x^4 ($\frac{1}{60} \cdot 24 = \frac{2}{5}$). It is not exact for x^6, for which it gives 0 instead of $\frac{2}{7}$, so the degree of precision is exactly 5.

4. **Degree of precision.** The degree of precision of the following quadrature rule is 3:

$$\int_{-1}^{1} f(x)\,\mathrm{d}x \approx \frac{4}{3}f\left(-\frac{1}{2}\right) - \frac{2}{3}f(0) + \frac{4}{3}f\left(\frac{1}{2}\right).$$

True. The rule is exact for 1 ($\frac{4}{3} - \frac{2}{3} + \frac{4}{3} = 2$) and for x^2 ($\frac{4}{3} \cdot \frac{1}{4} \cdot 2 = \frac{2}{3}$), and by symmetry for x and x^3. It is not exact for x^4: it gives $\frac{1}{6}$ instead of $\frac{2}{5}$.

5. **Degree of precision.** For any natural number $N > 0$ and distinct nodes $x_0 < \ldots < x_N$, there exist weights $w_0, \ldots, w_N$ such that the quadrature rule

$$\int_{-1}^{1} u(x)\,\mathrm{d}x \approx \sum_{n=0}^{N} w_n u(x_n)$$

has a degree of precision at least equal to N.

True. Take $w_n = \int_{-1}^{1} \ell_n(x)\,\mathrm{d}x$, where ℓ_n are the Lagrange polynomials. The rule then integrates exactly the interpolant of u, which is equal to u for every $u \in \mathcal{P}_N$.

6. **Degree of precision.** The closed Newton–Cotes rule with 3 equally spaced points (Simpson's rule) is exact for all polynomials of degree up to 3.

True. Simpson's rule is exact for quadratics by construction and, by symmetry, also for x^3.

7. **Composite quadrature.** Simpson's composite integration rule is exact for every polynomial f, whatever its degree.

False. The composite rule is exact only for polynomials of degree at most 3. For instance, it is not exact for x^4.

8. **Composite quadrature.** With $h = \frac{b-a}{n}$ and $x_i = a + ih$, the following formula is the composite version, with n cells, of the quadrature rule (1):

$$\int_{a}^{b} f(x)\,\mathrm{d}x \approx \frac{h}{2}\sum_{i=0}^{n-1}\left(f\left(x_i + \frac{h}{3}\right) + f\left(x_i + \frac{2h}{3}\right)\right). \tag{2}$$

True. Mapping $[-1, 1]$ to the cell $[x_i, x_{i+1}]$ sends the nodes $\pm\frac{1}{3}$ to $x_i + \frac{h}{2} \pm \frac{h}{6}$, that is, to $x_i + \frac{h}{3}$ and $x_i + \frac{2h}{3}$, and the weights 1 become $\frac{h}{2}$.

9. **Convergence estimate.** Suppose that $f \in C^\infty[a,b]$ and let $I_n[f]$ denote the approximate integral of f using the composite trapezoidal rule with $n+1$ equidistant integration points. Then there exists $C \in (0,\infty)$ such that

$$\forall n \in \{1,2,\dots\}\,, \qquad \Big|I[f] - I_n[f]\Big| \leqslant \frac{C}{n^2}\,, \qquad I[f] := \int_a^b f(x)\,\mathrm{d}x\,.$$

True. The error is bounded by $\frac{(b-a)^3}{12}\frac{\max|f''|}{n^2}$.

10. **Convergence estimate.** With the notation of the previous item, it holds for every $f \in C^\infty[a,b]$ that

$$\lim_{n\to\infty} n^2\Big|I[f] - I_n[f]\Big| = 0\,.$$

False. The order 2 is sharp in general. For $f(x) = x^2$ on $[0,1]$ the error is exactly $\frac{1}{6n^2}$, so $n^2|I[f] - I_n[f]| = \frac{1}{6}$ for all n.

11. **Convergence estimate.** Suppose that $f \in C^\infty[a,b]$ and let $I_n[f]$ denote the approximate integral of f using the composite Simpson rule with $n+1$ equidistant integration points, n even. Then it holds that

$$\lim_{n\to\infty} n^2\Big|I[f] - I_n[f]\Big| = 0\,.$$

True. For $f \in C^4$ the error of the composite Simpson rule is $O(n^{-4})$, so $n^2|I[f] - I_n[f]| = O(n^{-2}) \to 0$.

12. **Convergence estimate.** Let $u(x) = (1+25x^2)^{-1}$ and let

$$I_N^T = \frac{1}{2N}\Big(u(x_0) + 2u(x_1) + \cdots + 2u(x_{N-1}) + u(x_N)\Big)\,, \qquad x_i = \frac{i}{N}\,,$$

be the composite trapezoidal approximation of $\int_0^1 u(x)\,\mathrm{d}x$. Then I_N^T diverges in the limit $N \to \infty$.

False. The function is continuous, hence Riemann integrable, and I_N^T is a Riemann sum that converges to the integral.

13. **Convergence estimate.** Let $I_n[f]$ denote the right-hand side of the composite formula (2), with n cells. Then for every $f \in C^\infty[a,b]$ it holds that

$$\lim_{n\to\infty} n^3\Big|I[f] - I_n[f]\Big| = 0, \qquad I[f] := \int_a^b f(x)\,\mathrm{d}x\,.$$

False. The rule has degree of precision 1, so the error is of order $h^2 = O(n^{-2})$, and this is sharp. For $f(x) = x^2$ on $[0,1]$, each cell contributes an error $\frac{h^3}{18}$, so $|I[f] - I_n[f]| = \frac{1}{18n^2}$ and $n^3|I[f] - I_n[f]| = \frac{n}{18} \to \infty$.

14. **Julia.** The following code implements the composite version, with n cells, of the quadrature rule (1):

```
function I_approx(f, a, b, n)
    x = LinRange(a, b, n + 1)
    h = (b - a) / n
    m = x[1:n] .+ h/2
    return h/2 * sum(f, m .- h/6) + h/2 * sum(f, m .+ h/6)
end
```

True. On the cell with midpoint m and width h, the change of variable $x = m + \frac{h}{2}t$ maps the rule on $[-1,1]$ to $\frac{h}{2}\big(f(m - \frac{h}{6}) + f(m + \frac{h}{6})\big)$. The code sums this over the n cells, with midpoints `m`.

15. **Julia.** The following code correctly implements the composite Simpson rule with n cells, where n is even:

```
function I_approx(f, a, b, n)
    x = LinRange(a, b, n + 1)
    h = (b - a) / n
    return h/3 * (f(x[1]) + 2sum(f, x[2:2:end-1])
                  + 4sum(f, x[3:2:end-2]) + f(x[end]))
end
```

False. The weights are swapped. In Julia, `x[2:2:end-1]` are the nodes $x_1, x_3, \dots, x_{n-1}$ (in the 0-based numbering of the notes), which must get weight $\frac{4h}{3}$, whereas `x[3:2:end-2]` are $x_2, x_4, \dots, x_{n-2}$, which must get weight $\frac{2h}{3}$. Here it is the opposite.

16. **Bonus.** There exist nodes x_1, x_2, x_3 and weights w_1, w_2, w_3 such that the rule

$$\int_{-1}^{1} u(x)\,\mathrm{d}x \approx w_1 u(x_1) + w_2 u(x_2) + w_3 u(x_3)$$

has a degree of precision equal to 12. *Justify briefly.*

False. Suppose the degree of precision were 12 (in particular at least 6) and set $q(x) = (x-x_1)^2(x-x_2)^2(x-x_3)^2 \in \mathcal{P}_6$. Then the rule gives 0 while $\int_{-1}^{1} q\,\mathrm{d}x > 0$, a contradiction. In general, n nodes yield a degree of precision at most $2n-1$, here 5.